Supervised Machine Learning and Classification

Learning Outcomes

By completing these activities, you will be able to:

- Identify supervised learning problems.
- Distinguish between classification and regression tasks.
- Analyse the bias-variance trade-off.
- Explain why no single algorithm is best for every problem.

Activity 1: Identifying Supervised Learning Problems

For each scenario:

1. Decide whether it can be framed as a **supervised learning** problem.
2. Identify the:
    - Inputs (features)
    - Outputs (labels)
3. State whether it is:
    - Classification
    - Regression

Scenarios

- Medical diagnosis and treatment recommendation
- Customer support chatbot
- Offensive content detection
- Chess-playing system
- Ship detection in satellite images
- Movie recommendation system

Activity 2: Bias-Variance Analysis

For each change, predict whether:

- Bias increases or decreases

- Variance increases or decreases

Changes

- Add more training data
- Add more features
- Use a more complex model
- Train longer
- Remove outliers
- Reduce regularisation

Activity 3: No Free Lunch Theorem

Read the paper by Wolpert & Macready (1997).

D. H. Wolpert and W. G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.

Tasks

1. Explain the No Free Lunch theorem.
2. Why is there no universally best algorithm?
3. Give a real-world example.
4. Discuss implications for model selection.